

Project Synopsis Template

Automated Water Quality Evaluation using Decision Trees and Neural Networks

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
ANNAMDASU THANUJA	NMREC	ECE	9346259067	thanujaannamdasu@gmail.com	LEAD
INDRAKANTI KRISHNAVENI	NMREC	ECE	9347160896	krishnaveniindrakanti6@gmail.com	MEMBER
ANNAMDASU ASRITHA	NMREC	ECE	8686444126	Annamdasuasritha03@gmail.com	MEMBER
KURA SIRISREE	NMREC	ECE	8328106067	sirisreekura@gmail	MEMBER

Program Track and Domain

Program Track (Eng-HUB / Eng-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Eng-SPOKE	Machine Learning and Deep Learning

Guide Details

Mr. MANOJ KUMAR

- Problem Statement

Water contamination is a major environmental and health issue caused by industrial waste, chemicals, and pollutants. Traditional water quality testing methods are time-consuming and require laboratory analysis. This project aims to develop an automated water potability evaluation system using Decision Tree and Neural Network algorithms to predict whether water is safe for drinking based on water quality parameters. The system provides fast, accurate, and efficient prediction through a Streamlit application.

Objectives :

- To collect and preprocess the water potability dataset.
- To analyze parameters such as pH, hardness, solids, conductivity, and turbidity.
- To implement Decision Tree and Neural Network models for prediction.
- To compare the performance and accuracy of both models.
- To build an interactive water quality prediction system using Streamlit.

4. Methodology / Implementation Plan

1. Data Collection

- Collect Water Potability dataset from Kaggle.

2.Data Preprocessing

- Handle missing values
- Remove duplicates
- Normalize dataset

3. Exploratory Data Analysis (EDA)

- Analyze feature distributions
- Visualize correlations and patterns

4.Model Building

- Decision Tree Classifier
- Artificial Neural Network (ANN)

5.Model Evaluation

- Accuracy
- Precision
- Recall
- Confusion Matrix

6.Deployment

- Develop user interface using Streamlit
- Deploy application locally

Algorithms / Models Used

- Decision Tree Classifier
- Artificial Neural Network (ANN)
- Classification Algorithms
- DataNormalization Techniques

Tools & Technologies

- Jupyter Notebook
- Streamlit
- Python
- Machine Learning
- Neural Networks

• Resources and Timeline

Resources

- Python libraries: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, Matplotlib
- Dataset: Water Potability dataset (Kaggle or CSV file)
- Tools: Jupyter Notebook / VS Code, Streamlit
- Deployment: Local host / Streamlit Cloud
- Hardware: Basic laptop (no GPU required for small models)

Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Data collection and preprocessing (handling missing values, normalization)	Week 1
Phase 2	Exploratory Data Analysis (EDA) and visualization	Week 2
Phase 3	Train Decision Tree model	Week 3
Phase 4	Train Neural Network model and performance comparison	Week 4
Phase 5	Train Neural Network model and performance comparison	Week 5
Phase 6	Testing, debugging, and final deployment	Week 6
Phase 7	Final documentation	Week 7

• Expected Outcomes / Conclusion

By the end of the project:

- The system will successfully predict water potability using machine learning models like Decision Tree and Neural Network.
- The model will help identify whether water is safe or unsafe for drinking based on input water quality parameters.
- A Streamlit web application will provide an easy-to-use interface for real-time predictions.
- Comparison of models will show which algorithm gives higher accuracy and better performance.
- This project demonstrates how AI/ML techniques can be used to support public health and water safety monitoring.

• References

- Kaggle Water Potability Dataset
<https://www.kaggle.com/datasets/adityakadiwal/water-potability>
- Scikit-learn Documentation
<https://scikit-learn.org/stable/>
- TensorFlow Documentation
<https://www.tensorflow.org/>
- Keras Documentation
<https://keras.io/>
- Pandas Documentation
<https://pandas.pydata.org/docs/>
- NumPy Documentation
<https://numpy.org/doc/>
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- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow
<https://www.oreilly.com/library/view/hands-on-machine-learning/9781492032632/>
- Data Mining: Concepts and Techniques
<https://www.sciencedirect.com/book/9780123814791/data-mining-concepts-and-techniques>
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